FTP Credential Interception

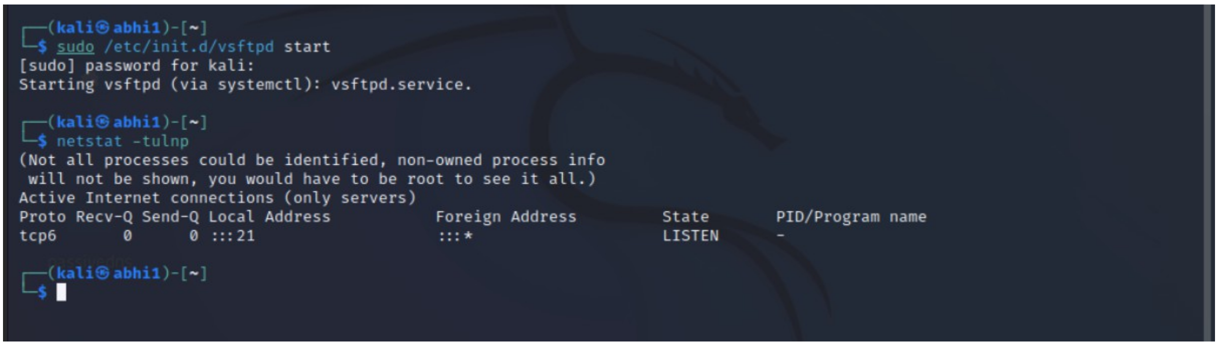

Fig 1.1

In the above figure 1.1 we are initialising a directory (/etc/init.d) containing the scripts For managing the services and by saying START we are starting a **very secure FTP DAEMON server which allows the system as an FTP server.**

In the second line we use the command **netstat** for displaying the network statistics Tables etc..

-tulnp are some set of connections.

-t denotes the display of **TCP connections**
-u denotes the display of **UDP connections**
-l denotes the display of listening sockets where the system is expecting the incoming connections
-n denotes the numerical addresses for quick and clean outputs
-p denotes process id

Now by executing these it shows all the present listening network ports and starts listening to the FTP incoming sessions so as we can see that the local address is allotted with 21 port which is the default port for that server.

```
abhishek@abhi2: ~ (on abhi2)
File  Actions  Edit  View  Help
debug          glob           mls            passive        reget          status
ftp> cd textfiles
250 Directory successfully changed.
ftp> pwd
Remote directory: /home/abhi/textfiles
ftp> ls
229 Entering Extended Passive Mode (|||53642|)
150 Here comes the directory listing.
-rw-r--r--    1 1002     1002          101 Oct 08 03:21 ABAs.txt
226 Directory send OK.
ftp> help
Commands may be abbreviated.  Commands are:

!              delete         hash           mlsd           pdir           remopts        struct
$              dir            help           mlst           pls            rename         sunique
account        disconnect     idle           mode           pmlsd          reset          system
append         edit           image          modtime        preserve       restart        tenex
ascii          epsv           lcd            more           progress       rhelp          throttle
bell           epsv4          less           mput           prompt         rmdir          trace
binary         epsv6          lpage          mreget         proxy          rstatus        type
bye            exit           lpwd           msend          put            runique        umask
case           features       ls             newer          pwd            send           unset
cd             fget           macdef         nlist          quit           sendport       usage
cdup           form           mdelete        nmap           quote          set            user
chmod          ftp            mdir           ntrans         rate           site           verbose
close          gate           mget           open           rcvbuf         size           xferbuf
cr             get            mkdir          page           recv           sndbuf         ?
debug          glob           mls            passive        reget          status
ftp> pwd
Remote directory: /home/abhi/textfiles
ftp> dir
229 Entering Extended Passive Mode (|||42223|)
150 Here comes the directory listing.
-rw-r--r--    1 1002     1002          101 Oct 08 03:21 ABAs.txt
226 Directory send OK.
ftp> cd txtfiles
```

Fig 1.2

Now we are setting up an FTP server on a kali 2 machine using TCP dump and analysing the packet capture also using wireshark.
The command cd txtfiles to change the present directory to txtfiles based on the given name of the file (ABAs.txt).

Pwd stands for print working directory in this case it is /home/abhi/textfiles in the remote FTP server.

To list the contents (ABAs.txt) we use dir command with permission of -rw-r -- r-- ,this means that the owner now has read and write permission while the others only have read access.

The owner in the sense the machine creates a unique user identifier number (1002) and the group identifier number is also 1002 with a file size of 101 bytes with respective time frames in ABas.txt file.

Now we are entering into the 229 extended passive mode to establish connection to the server with a port number 42321 for data connection.

```
abhishek@abhi2: ~ (on abhi2)
File  Actions  Edit  View  Help
close          gate           mget           open           rcvbuf         size           xferbuf
cr             get            mkdir          page           recv           sndbuf         ?
debug          glob           mls            passive        reget          status
ftp> pwd
Remote directory: /home/abhi/textfiles
ftp> dir
229 Entering Extended Passive Mode (|||42223|)
150 Here comes the directory listing.
-rw-r--r--    1 1002     1002          101 Oct 08 03:21 ABAs.txt
226 Directory send OK.
ftp> cd txtfiles
550 Failed to change directory.
ftp> cd textfiles
550 Failed to change directory.
ftp> dir
229 Entering Extended Passive Mode (|||41630|)
150 Here comes the directory listing.
-rw-r--r--    1 1002     1002          101 Oct 08 03:21 ABAs.txt
226 Directory send OK.
ftp>
ftp>
ftp> get ABAs.txt
local: ABAs.txt remote: ABAs.txt
229 Entering Extended Passive Mode (|||11112|)
150 Opening BINARY mode data connection for ABAs.txt (101 bytes).
100% |********************************************************************************|   101        1.66 MiB/s    00:00 ETA
226 Transfer complete.
101 bytes received in 00:00 (38.02 KiB/s)
ftp>
ftp>
ftp>
ftp> quit
221 Goodbye.

┌──(abhishek㉿abhi2)-[~]
└─$
```

Fig 1.3

Now again in Fig 1.3 we are reentering passive mode again with different 14630 port for the connection and transmission. Each time we enter into a passive mode a new port is generated. Now we use get ABAs.txt to retrieve from the server with 150 opening binary mode data connection for the .txt file with a file size of 101 bytes. 226 the transfer is completed and has transferred from the client to the respective server with a speed of 1.66 Mib/s
We are ending the FTP session with a command 221 goodbye.

```
┌──(kali㉿abhi1)-[~/task2]
└─$ sudo tcpdump -n -r task2.pcap tcp -x
reading from file task2.pcap, link-type EN10MB (Ethernet), snapshot length 262144
03:51:22.190988 IP 192.168.2.102.52076 > 192.168.2.101.21: Flags [S], seq 968048151, win 65535, options [mss 1460,sackOK,TS
val 1509659868 ecr 0,nop,wscale 2], length 0
        0x0000:  4500 003c 7aa4 4000 4006 39fc c0a8 0266
        0x0010:  c0a8 0265 cb6c 0015 39b3 3e17 0000 0000
        0x0020:  a002 ffff 8fc5 0000 0204 05b4 0402 080a
        0x0030:  59fb 94dc 0000 0000 0103 0302
03:51:22.191062 IP 192.168.2.101.21 > 192.168.2.102.52076: Flags [S.], seq 497401783, ack 968048152, win 65160, options [mss
 1460,sackOK,TS val 1865778948 ecr 1509659868,nop,wscale 7], length 0
        0x0000:  4500 003c 0000 4000 4006 b4a0 c0a8 0265
        0x0010:  c0a8 0266 0015 cb6c 1da5 bfb7 39b3 3e18
        0x0020:  a012 fe88 864a 0000 0204 05b4 0402 080a
        0x0030:  6f35 8704 59fb 94dc 0103 0307
03:51:22.192538 IP 192.168.2.102.52076 > 192.168.2.101.21: Flags [.], ack 1, win 16384, options [nop,nop,TS val 1509659870 e
cr 1865778948], length 0
        0x0000:  4500 0034 7aa5 4000 4006 3a03 c0a8 0266
        0x0010:  c0a8 0265 cb6c 0015 39b3 3e18 1da5 bfb8
        0x0020:  8010 4000 aae2 0000 0101 080a 59fb 94de
        0x0030:  6f35 8704
03:51:22.199512 IP 192.168.2.101.21 > 192.168.2.102.52076: Flags [P.], seq 1:21, ack 1, win 510, options [nop,nop,TS val 186
5778957 ecr 1509659870], length 20: FTP: 220 (vsFTPd 3.0.3)
        0x0000:  4500 0048 0af9 4000 4006 a99b c0a8 0265
        0x0010:  c0a8 0266 0015 cb6c 1da5 bfb8 39b3 3e18
```

Fig 1.4

We are analysing the FTP traffic between kali 1 and kali 2 by using tcpdump to read the task.pcap and capturing with port number 52076 for kali 2 and port number 21 for kali 1 with a seq number of 497401783 and acknowledgement number of 968948152.

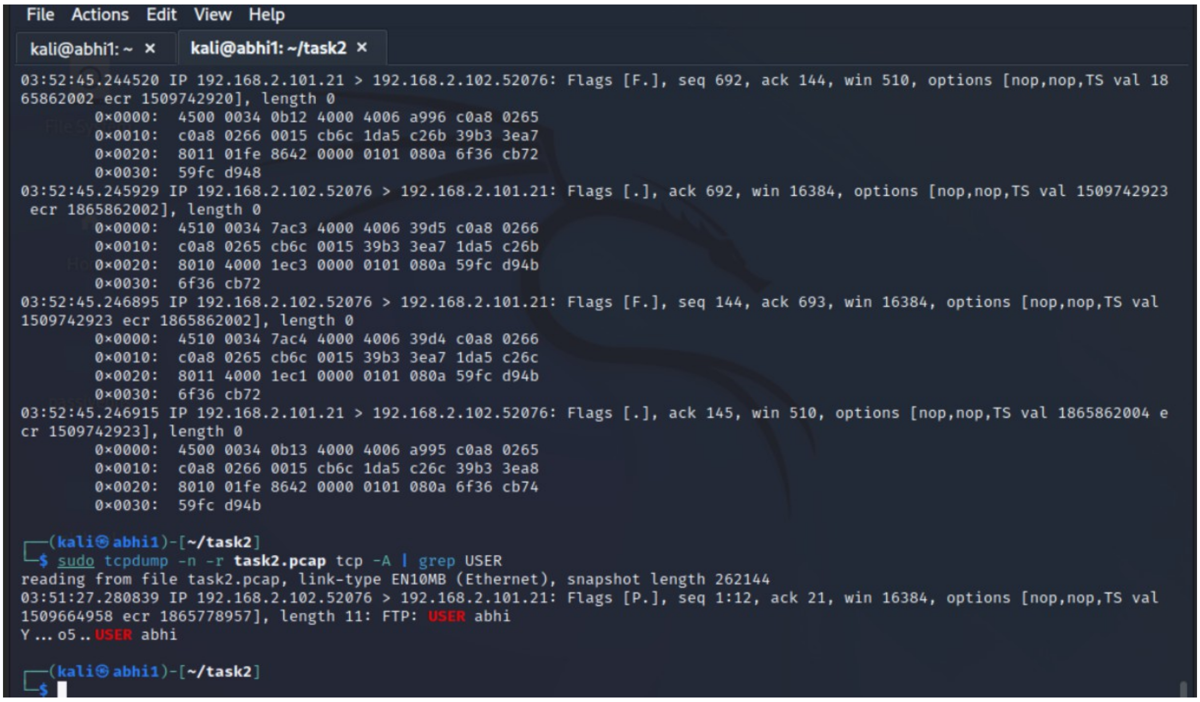

Fig 1.5

To find the username we use the **grep command** which is used to log into the FTP server.In this case it is **abhi and password is kali** these credentials were transmitted in a very plain text which makes the session very weak and vulnerable.

-A stands for printing the packets in ASCII format for easy understanding.

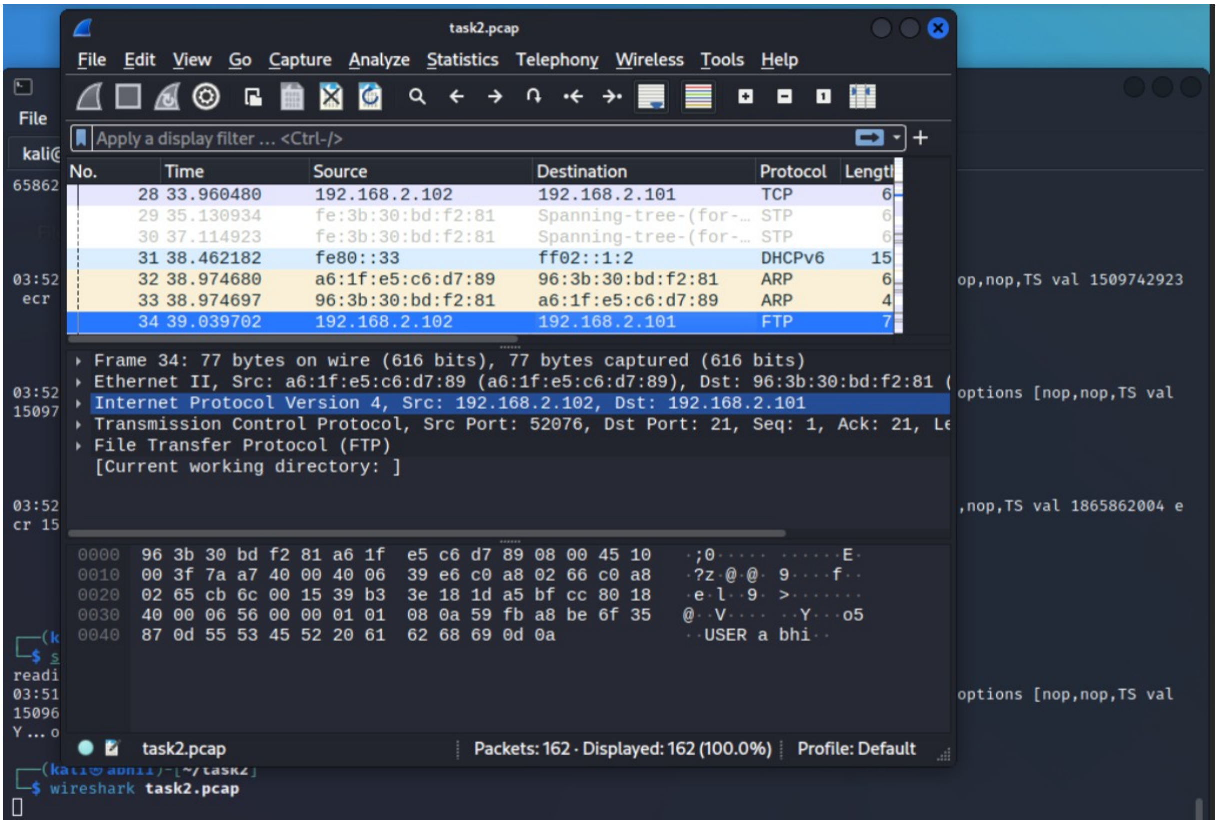

Fig 1.6

We have started the wireshark analysis to complete the FTP session, so in task2.pcap file display we can see the source and destination Ip4v with TCP source port as 52076 and destination port of 21. we got the user as abhi in the **figure 2.6**

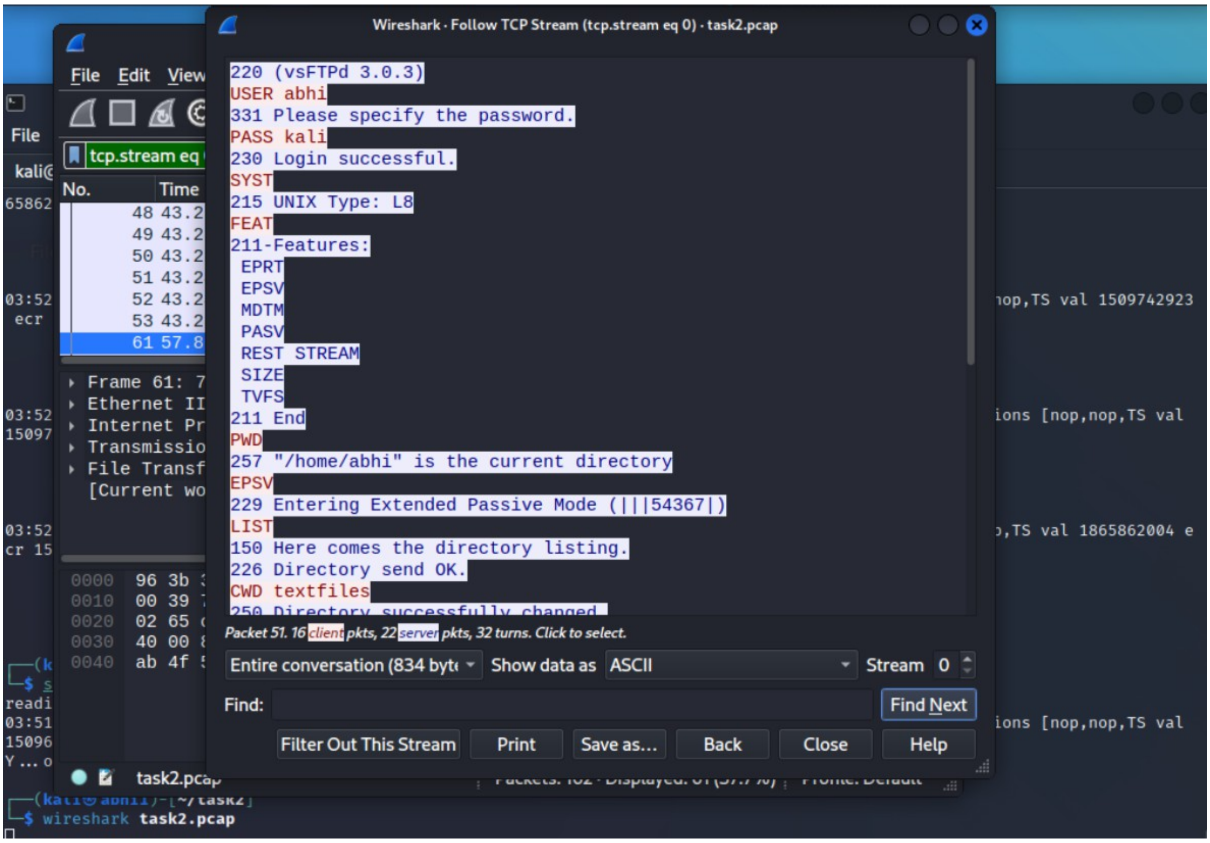

Fig 1.7

In figure 1.7 we can see the user and password details in plain text in the working directory of /home/abhi/ and entering extended passive mode (|| |54367|) we can see 150 opening port BINARY mode data and data connection and 226 means the transfer has completed.

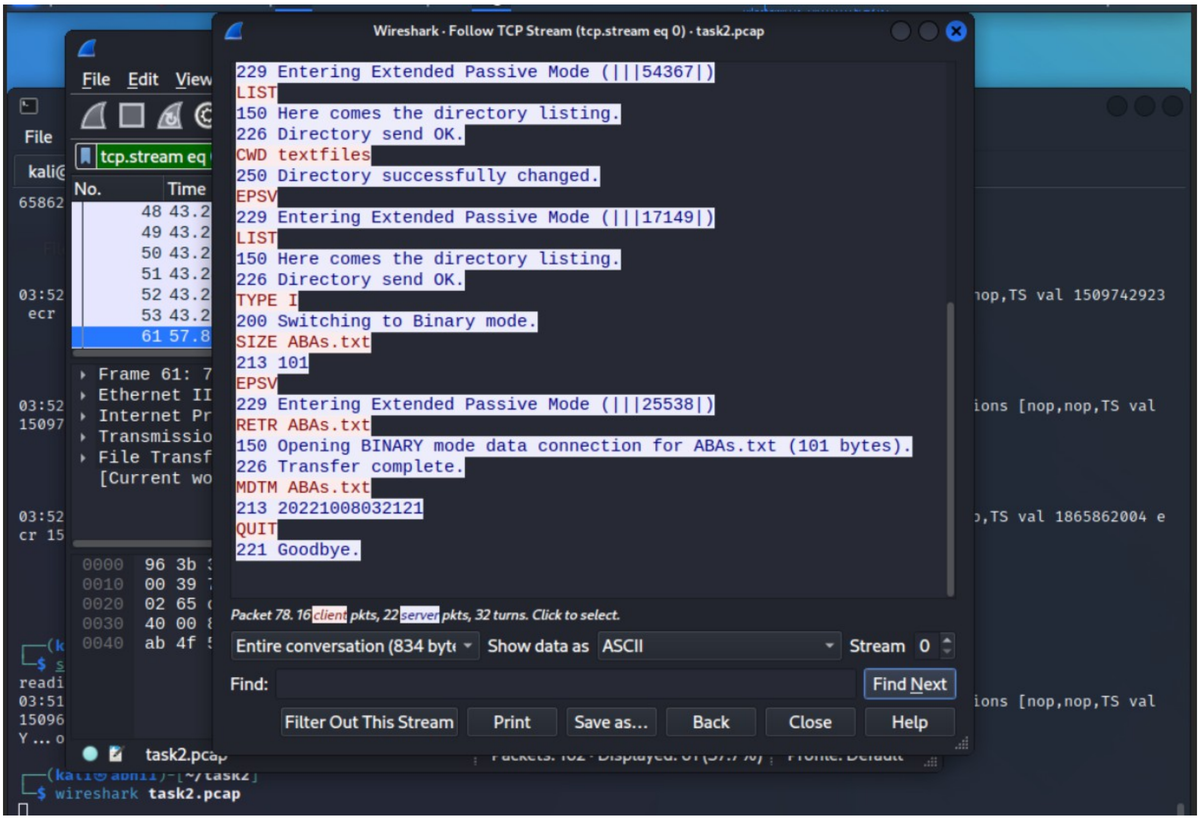

Fig 1.8

In figure 1.8 we can see the user and password details in plain text in the working directory of /home/abhi/ and entering extended passive mode (|| |25538|) we can see 150 opening port BINARY mode data and data connection,the command RETR ABAs.txt was for start downloading the file and 226 means the transfer has completed 221 quit.